

System: NorthStar Omnichannel Payments Platform

Classification: Restricted

Deployment: Hybrid cloud edge plus on-prem restricted zone

Regulatory Scope: OSFI B-13, PCI DSS, PIPEDA, FINTRAC

Trust Boundary: Customer and Partner Edge

Contains:

- Consumer Mobile App [external_entity]
- Open Banking Partner [external_entity]
- Treasury Portal User [external_entity]
- API Gateway [process]

Trust Boundary: Payments Control Plane

Contains:

- Payments Orchestrator [process]
- Fraud Scoring Engine [process]
- AML Screening Service [process]
- SWIFT Connector [process]

Trust Boundary: Restricted Data Zone

Contains:

- Core Banking Ledger [data_store]
- Card Token Vault [data_store]
- Case Management Store [data_store]

Data Flows:

- Consumer Mobile App -> API Gateway: OAuth2 + payment initiation
- Open Banking Partner -> API Gateway: mTLS signed payment request
- Treasury Portal User -> API Gateway: privileged batch approval
- API Gateway -> Payments Orchestrator: normalized payment instruction
- Payments Orchestrator -> Fraud Scoring Engine: transaction scoring request
- Payments Orchestrator -> AML Screening Service: sanctions screening request
- Payments Orchestrator -> SWIFT Connector: signed SWIFT payment message
- Payments Orchestrator -> Core Banking Ledger: posting and balance update
- Payments Orchestrator -> Card Token Vault: tokenization lookup
- Fraud Scoring Engine -> Case Management Store: investigation case record
- Payments Orchestrator -> Open Banking Partner: payment status callback

Security Properties:

- API Gateway: internet_facing=true, uses_auth=true, validates_input=true, uses_encryption=true
- Payments Orchestrator: handles_sensitive_data=true, uses_encryption=true
- Core Banking Ledger: encrypted_at_rest=true, has_backup=true
- Card Token Vault: stores_credentials=true, encrypted_at_rest=true
- Open Banking Partner: authenticated=true, trusted=true